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Shortest Cable Routing of Heliostats in Solar Tower Power Plants

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Aachen, im Juli 2018

Tran ► [sign lab course report before handing in]

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1 Introduction

In this work we present two heuristics that compute valid solutions to an optimization problem regarding a cabling problem in solar tower power plants. Such a plant focuses sunlight with the help of heliostats onto a receiver that is installed at the top of the solar tower. This energy heats up a medium like oil or sodium in order to power a turbine which generates electric energy in return. Such a plant consists of a solar tower and heliostats that have to be connected to the solar tower with two types of cables, namely a power and data cable. Both cables are needed at each heliostat as the sun's position has to be taken into account for an efficient exploitation of solar energy. Regarding the cabling we have several limitations with respect to connectivity and planarity constraints. First, cables have restrictions regarding their maximum lengths between heliostats. Moreover, there are capacity constraints for each cable, namely that cables can only connect so many heliostats. Second, the planarity constraint is enforced in order to have a low maintenance cost when digging up trenches.

Another source of energy is utilized in offshore wind farms for which the cabling problem has been studied in [5, 2, 4]. The same constraints like planarity are also present in their cabling problem. Our work builds up on two bachelor's theses that studied the solar tower power plants [6, 10].

In the following we will give a formal definition of our problem and present two heuristics that apply the local search optimization method [1, 11]. This approach was chosen due to the high complexity of the cabling problem, which is especially difficult to solve for large instances that inhibit a lot of heliostats. Then, an evaluation part is followed where we have applied both heuristics onto the PS10 solar tower power plant [7, 8]. Afterwards we give a short summary and possible future work.

2 Problem statement

Formally, we are given an undirected graph $G = (V, E)$ where $V = \{T\} \cup V_H$ with T being our solar tower and $V_H := \{v_1, \dots, v_n\}$ denotes the set of heliostats. Note that we work in a metric space $(\mathbb{R}^2, d)$, thus we denote with $v_x \in \mathbb{R}$ and $v_y \in \mathbb{R}$ both coordinates for each vertex $v \in V$. Moreover, the metric is simply the euclidean distance

$$d(v, w) := \sqrt{(v_x - w_x)^2 + (v_y - w_y)^2}$$

between nodes $v, w \in V$. In other words, each vertex $v \in V$ has a specified position in a 2D plane.

Before a cable can be installed a trench has to be constructed by workers for which we have to pay installation costs. Let us denote these costs per meter with $c^{\text{trench}} \in \mathbb{R}^+$. In other words, the total costs of providing a trench between $v, w \in V$ is $c_{vw}^{\text{trench}} := d(v, w) \cdot c^{\text{trench}}$.

2.1 Data cable and local control units

We can install different types of cables which have length and connection restrictions depending on its type. For this matter let D be the set of data cables with costs $c_d^{\text{data}} \in \mathbb{R}^+$ per meter for each cable $d \in D$. We further define for readability the costs $c_{d,vw}^{\text{data}} := d(v, w) \cdot c_d^{\text{data}}$ of installing a cable of type d on the edge $vw \in E$. Furthermore, each data cable $d \in D$ has a length restriction $L_d^{\text{data}} \in \mathbb{R}^+$ and a restriction on how many heliostats it can connect to, that is, $M_d^{\text{data}} \in \mathbb{N}$ for $d \in D$.

Aside from the cables we have to build *local control units* (LCU) at each heliostat. These LCUs are used for communication processing and may allow branching by e.g. installing an 8- or 16-port LCU at an heliostat. Each of these LCU has connectivity-restrictions for each cable type $d \in D$. For this matter, let S the set of all LCUs where each $s \in S$ has installation costs $c_s^{\text{lcu}} \in \mathbb{R}^+$. The connectivity or vertex-degree restrictions are given by $l_{s,d}, u_{s,d} \in \mathbb{N}$ for each LCU $s \in S$ and cable type $d \in D$. Both restrictions state that the LCU $s \in S$ has to connect at least $l_{s,d}$ cables of type $d \in D$, whereas $u_{s,d}$ is the upper bound of incident cables of type $d \in D$ at an heliostat that installed LCU $s \in S$.

2.2 Power cables

► [formal definition]

2.3 Cabling problem

► [tbox with formal problem and optimization criterion]

► [capacity constraint]

3 Data

We present real world data regarding costs per meter for various data and power cables. Moreover, many different LCUs as well as trench costs are provided for which we can test our heuristics on. In particular we have also coordinates of heliostats from the PS10 solar tower power plant [7, 8], a sketch of this instance can be seen in fig. 1.

3.1 Trench costs

We consider four different countries that are possible locations for our solar tower plant. Each trench has to be constructed by workers and has a depth of 1m as well as 1m in width and length.

Figure 1: PS10 instance with its heliostats and the solar tower at coordinate $(0, 0)$.

Table 1: Trench costs for different countries

Country	Costs [€/m]
Australia	50
Spain	25
South Africa	10
United Arab Emirates	10

3.2 Data Cable

Regarding the data cable we are able to install either a copper or fiberglass cable.

Table 2: The copper cable is much cheaper however it has harsh restrictions regarding its usage. The fiberglass cable has actually no maximum connection count restriction. We use a restriction of 128 heliostats in order to comply to certain protocol usage.

Data cable type	Costs [€/m]	Length restriction [m]	Max. connection count
Copper	0.7	100	2
Fiberglass	2	∞	128

► [large constant for length restriction for fiberglass?]

3.2.1 Local Control Units

Exactly one local control units is installed at every heliostat so that communication is enabled.

Table 3: Local control units with their degree and cable restrictions for both cable types.

LCU type s	Costs [€]	Copper		Fiberglass	
		$l_{s,COP}$	$u_{s,COP}$	$l_{s,FIG}$	$u_{s,FIG}$
Endpoint	10	1	1	0	0
Conductor	100	0	0	2	2
8-port copper	800	0	7	1	2
8-port fiberglass	800	0	0	1	9
16-port copper	1500	0	15	1	2
16-port fiberglass	1500	0	0	1	17

► [image of some example LCUs]

3.3 Power Cable

► [data; protective foil are simply added to costs]

4 Heuristics

In the following both sections we present our heuristics that are based on the local-search optimization method [1, 11]. The basic structure of such an algorithm is shown in algorithm 1. Basically we first compute an initial solution from the input and afterwards improve it stepwise until some stopping criterion is met. In principle the stepwise improvements are performed locally to sol such that each step yields a valid solution to our optimization problem.

Algorithm 1 A general framework of a local search heuristic.

```
1: procedure LOCAL-SEARCH( $input$ )
2:    $init := \text{COMPUTE-INITIAL-SOLUTION}(input)$ 
3:    $sol := init$ 
4:   while stopping criterion not met do
5:      $x := \text{COMPUTE-NEXT-NEIGHBOR-SOLUTION}(sol)$ 
6:     if  $c(x) < c(sol)$  then
7:        $sol := x$ 
8:     end if
9:   end while
10:  return  $sol$ 
11: end procedure
```

One may interpret this by jumping from one point in the valid solution search space to another one in a certain local neighborhood. This interpretation is depicted in fig. 2 where s_0 is the initial solution that we improve in a stepwise-manner by selecting a neighboring solution s_1 in its neighborhood $N(s_0)$. Obviously s_1 has lower total costs than s_2 , however, depending on the neighborhood size $|N(s_0)|$ it may be a computationally demanding task. Therefore, while generating the neighboring solutions in $N(s_0)$ we terminate the search process as soon as we encounter a solution $s_1 \in N(s_0)$ with $c(s_1) < c(s_0)$. This search strategy is called *first-improvement* as we do not generate the best neighboring solution but are satisfied with any kind of improvement. This process is then continued until some stopping criterion is met. In particular, one may terminate after $i \in \mathbb{N}$ steps or in our case terminate as soon as $N(s_i)$ does not yield any improving solution, hence s_i is called *locally optimal*.

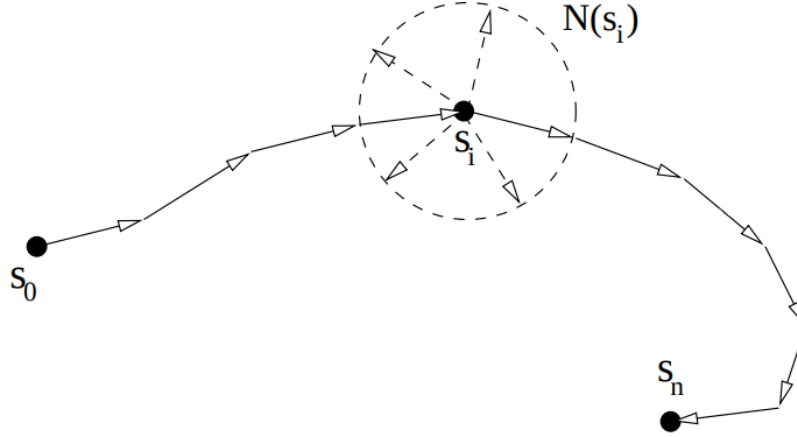


Figure 2: The search space where each point describes a valid solution to our optimization problem. We start from the initial solution s_0 improve it stepwise. For this matter we consider 'neighboring' solutions in $N(s_i)$ at the current solution s_i . When no neighboring solution is improving (i.e. has less total costs than current solution) we terminate our algorithm as s_n is a locally optimal solution.

► [both cable in one, assign power automatically]

4.1 Minimum Spanning Tree Approach

The fact that each heliostat has to be connected to the solar tower T with minimal costs infers utilizing a minimum spanning tree approach. We just need to consider the additional constraints of planarity as well as connectivity constraints. We thus choose Prim's algorithm [9, 3] to compute an initial, valid solution to our problem. For the local search improvement steps we choose the *cycle-neighborhood* function

$$N(T) := \{\tilde{T} \in \mathcal{T} \mid$$

5 Evaluation

► [Evaluation tables and so on]

6 Conclusion

► [Conclusion]

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